

1. (10 pts) Demonstrate the following 8-bit binary unsigned integer operation. Provide the corresponding operands and the results in decimal. Please show your work.

$$1010\ 0101 \times 10 =$$

binary

decimal

$$\begin{aligned} &(1)\ 0100\ 1010 \\ &2+8+64=74 \end{aligned}$$

$$\begin{aligned} &165 \times 2 = 330 \\ &330 \% 256 = 74 \end{aligned}$$

2. (10 pts) Convert the following 8-bit signed integers into decimal using Two's complement or convert the following decimal into 8-bit signed integers using Two's complement. Please show your work.

$$\begin{aligned} &1011\ 1000 \\ &=-2^7+2^5+2^4+2^3 = -128+32+16+8 = -72 \end{aligned}$$

$$\begin{aligned} &-85 \\ &85 = 64+16+4+1 = 2^6 + 2^4 + 2^2 + 2^0 = 01010101 \\ &-85 = 10101010+1 = 10101011 \end{aligned}$$

3. (8 pts) Convert the numbers into the specified number system.

1) Convert $(1023)_5$ to base-8 (Radix = 8);

$$1 \cdot 5^3 + 2 \cdot 5^1 + 3 = 138 \rightarrow (212)_8$$

2) Convert $(762)_8$ to base-5;

$$7 \cdot 8^2 + 6 \cdot 8 + 2 = 498 \rightarrow (3443)_5$$

4. (10 pts) Convert the following values (unsigned) to decimal. Please show your work.

$$0111.1011 = 7.6875$$

$$10101.101 = 21.625$$

5. (10 pts) Assume an 8-bit floating point format with a 4-bit exponent and a bias of 7 and implied 1. Provide the decimal representations of the following stored values.

Sign	Exponent				Mantissa		
1	1	1	0	0	1	0	1

$$-1.101 \cdot 2^5 = -110100 = -52$$

Sign	Exponent				Mantissa		
0	0	1	0	1	0	1	0

$$1.01 \cdot 2^{-2} = 0.0101 = 0.3125$$

6. (5 pts) What is the computational complexity of multiplying two $n \times n$ matrices using a naïve algorithm?

$$O(n^3)$$

7. (12 pts) Reformulate the following expression to deal with catastrophic cancellation and test the result using 3 digits of precision for $x = 0.01$.

$$\sqrt[4]{1+x} - \cos(x)$$

Exact Result (using calculator or computer)

$$0.00254$$

Reformulation

$$\begin{aligned} \sqrt[4]{1+x} - \cos(x) &= \frac{\sqrt[4]{1+x} - \cos^2 x}{\sqrt[4]{1+x} + \cos(x)} = \frac{\sqrt[4]{1+x} - 1}{\sqrt[4]{1+x} + \cos(x)} + \frac{\sin^2 x}{\sqrt[4]{1+x} + \cos(x)} \\ &= \frac{x}{(\sqrt[4]{1+x} + \cos(x))(\sqrt[4]{1+x} + 1)} + \frac{\sin^2 x}{\sqrt[4]{1+x} + \cos(x)} \end{aligned}$$

Original result

$$1.00 - 1.00 = 0$$

Reformulated result

$$\begin{aligned} &1.00, 1.00, 1.00 \\ &= x/4 + (\sin x)^2/2 = \\ &0.0025 + 0.00005 = 0.00255 \end{aligned}$$

Relative error
100%

Relative error
0.394%

8. (10 pts) Reformulate using the Mean Value Theorem and test the result using 3 digits of precision for $x = 0.01$. Show your work.

$$\sqrt{\pi + x} - \sqrt{\pi}$$

Exact Result (using calculator or computer)

0.00282

Reformulation

$$f(x) = x^{1/2} \quad f'(x) = 1/2 * x^{-1/2}$$

$$\rightarrow (\pi + x)^{1/2} - (\pi)^{1/2} = x^{1/2} * (\pi + x/2)^{-1/2} = x/2 * (\pi + x/2)^{-1/2}$$

Original result

Reformulated result

$$1.77 - 1.77 = 0$$

$$3.15^{1/2} = 1.77$$

$$1/1.77 = 0.565$$

$$0.005 * 0.565 = 0.00283$$

Relative error

Relative error

100%

0.355%

9. (10 pts) Use a Taylor series expansion to reformulate the following numerical problem. Keep terms up to and including $O(x^3)$. Calculate the relative error before and after reformulation for $x = 0.02$. Use 3 significant digits for each operation.

$$\frac{1 - \cos x}{x}$$

Exact Result (using calculator or computer)

0.00100

Reformulation

$$\cos(x) = \cos(0) - \sin(0) * x - \cos(0)/2 * x^2 + \sin(0)/6 * x^3$$

$$= 1 - x^2/2$$

$$\rightarrow x^2/2/x = x/2$$

Original result

0

Relative error

100%

Reformulated result

0.00100

Relative error

0%

10. (15 pts) Use synthetic division to compute $f(x) = e^{-x} \sin x^2$ for $x = 0.1$ with an error bound of $O(x^4)$ using 3 significant digits. Show this calculation using synthetic division.

$$e^x = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3$$

$$e^{-x} = 1 - x + \frac{1}{2}x^2 - \frac{1}{6}x^3$$

$$\sin x = \sin(0) + \cos(0)x$$

$$\sin x^2 = x^2$$

$$f(x) = x^2 - x^3$$

0.1	-1	1	0	0
	-0.1 0.09 0.00900			
	-1	0.9	0.09	0.00900